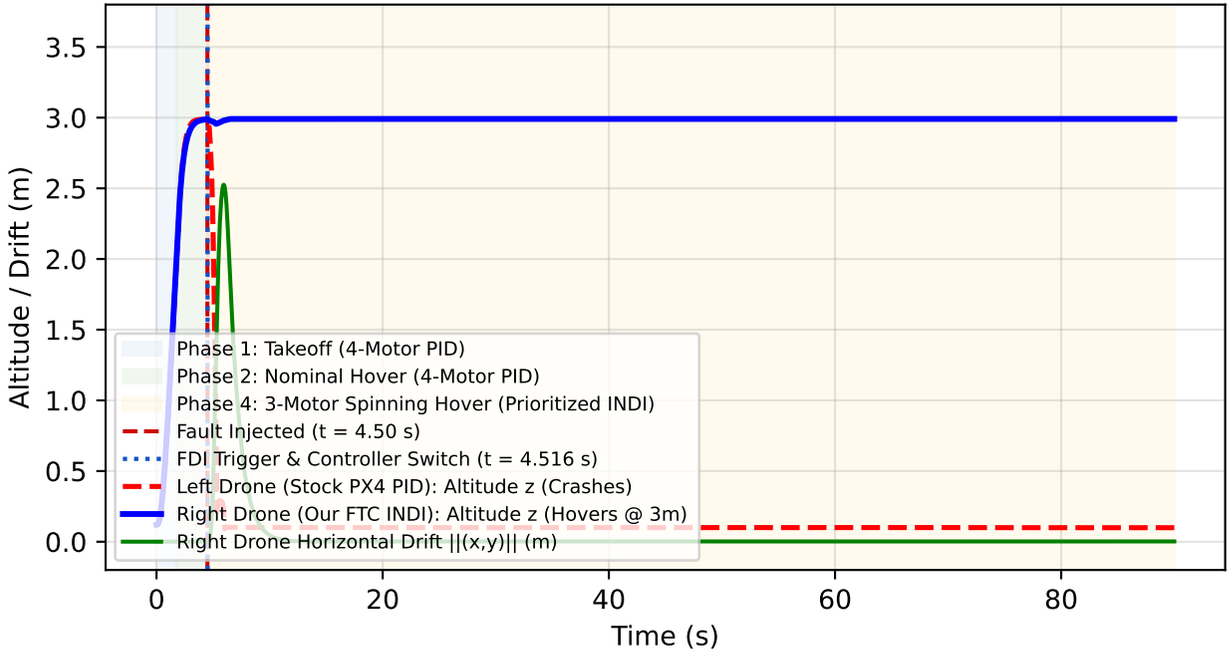
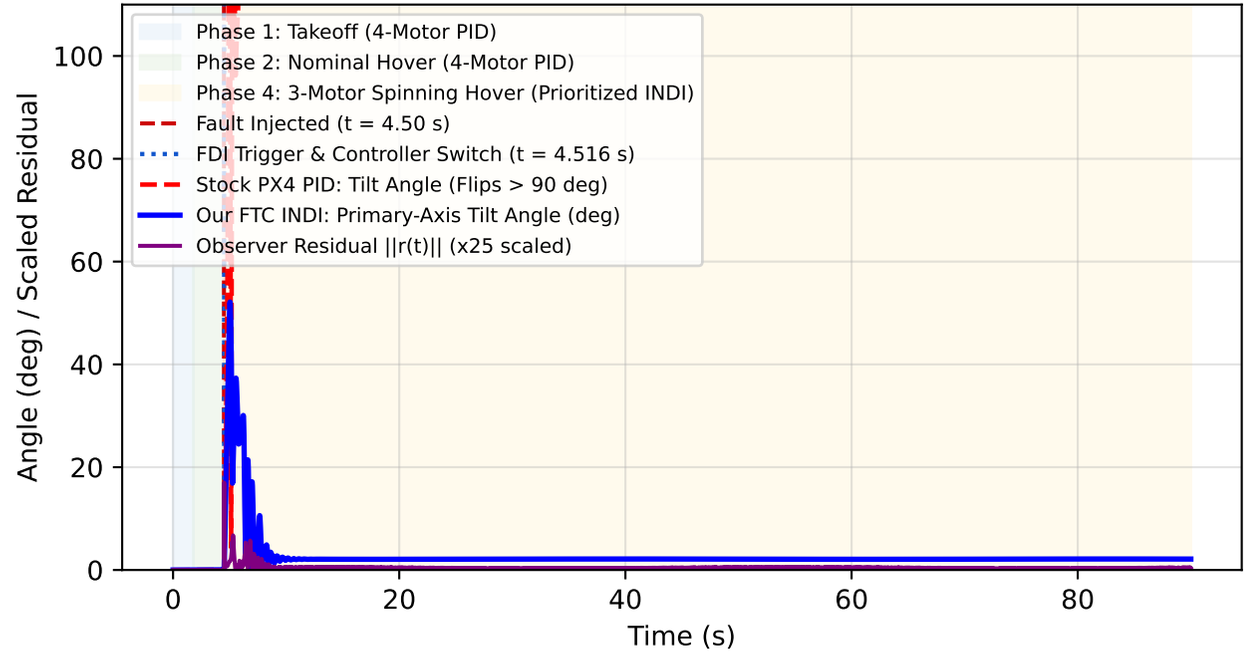
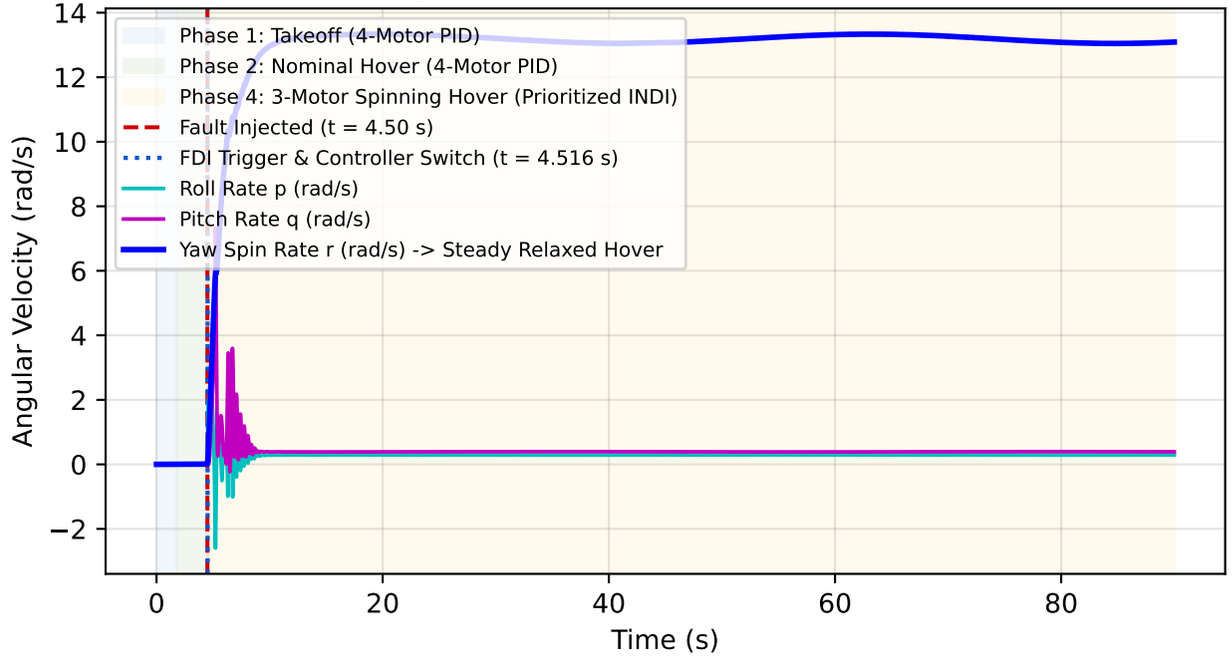
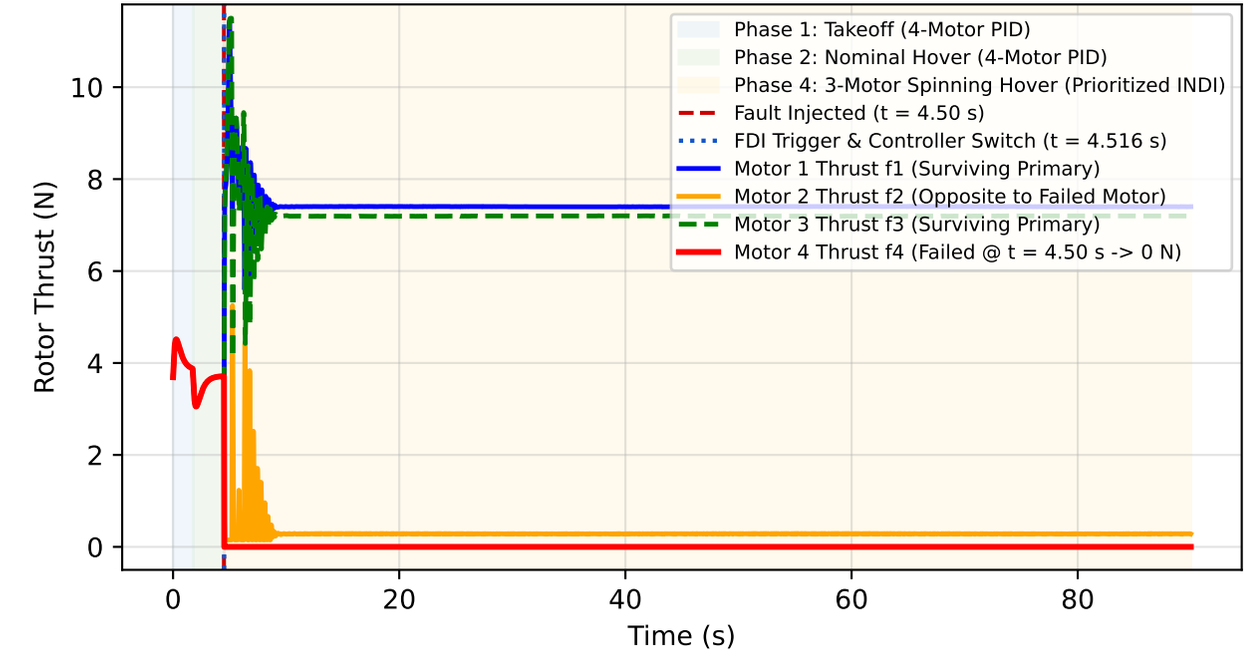

1. Altitude & Hover Recovery (Logged from live_3d_simulation.ulg)
Altitude / Drift (m)
Time (s)
Phase 1: Takeoff (4-Motor PID)
Phase 2: Nominal Hover (4-Motor PID)
Phase 4: 3-Motor Spinning Hover (Prioritized INDI)
Fault Injected (t = 4.50 s)
FDI Trigger & Controller Switch (t = 4.516 s)
Left Drone (Stock PX4 PID): Altitude z (Crashes)
Right Drone (Our FTC INDI): Altitude z (Hovers @ 3m)
Right Drone Horizontal Drift ||(x,y)|| (m)


2. Primary-Axis Tilt Angle & FDI Residual Spike
Angle (deg) / Scaled Residual
Time (s)
Phase 1: Takeoff (4-Motor PID)
Phase 2: Nominal Hover (4-Motor PID)
Phase 4: 3-Motor Spinning Hover (Prioritized INDI)
Fault Injected (t = 4.50 s)
FDI Trigger & Controller Switch (t = 4.516 s)
Stock PX4 PID: Tilt Angle (Flips > 90 deg)
Our FTC INDI: Primary-Axis Tilt Angle (deg)
Observer Residual ||r(t)|| (x25 scaled)


3. Body Angular Velocities Across Controller Switch
Angular Velocity (rad/s)
Time (s)
Phase 1: Takeoff (4-Motor PID)
Phase 2: Nominal Hover (4-Motor PID)
Phase 4: 3-Motor Spinning Hover (Prioritized INDI)
Fault Injected (t = 4.50 s)
FDI Trigger & Controller Switch (t = 4.516 s)
Roll Rate p (rad/s)
Pitch Rate q (rad/s)
Yaw Spin Rate r (rad/s) -> Steady Relaxed Hover


4. Rotor Thrust Reallocation: 4-Motor PID -> 3-Motor INDI
Rotor Thrust (N)
Time (s)
Phase 1: Takeoff (4-Motor PID)
Phase 2: Nominal Hover (4-Motor PID)
Phase 4: 3-Motor Spinning Hover (Prioritized INDI)
Fault Injected (t = 4.50 s)
FDI Trigger & Controller Switch (t = 4.516 s)
Motor 1 Thrust f1 (Surviving Primary)
Motor 2 Thrust f2 (Opposite to Failed Motor)
Motor 3 Thrust f3 (Surviving Primary)
Motor 4 Thrust f4 (Failed @ t = 4.50 s -> 0 N)